

Fantasy State and Forward XP Divergence

Forward Performance Validation Study

1. Fantasy State

Fantasy State classifies the relationship between a player's XP and the XP expected from his Overall Score. Expected XP is calculated as:

$$\text{Expected XP} = 813.86 + (47.78 \times \text{Overall})$$

XP Divergence is the percentage difference between actual XP and Expected XP. Players are classified as Strong ($> +5\%$), Stable (-2.5% to $+5\%$), Vulnerable (-5% to $< -2.5\%$), or High Risk ($< -5\%$). These states describe the current relationship between production and the underlying five-metric profile.

2. Research Design

A September 2, 2026 population snapshot was used as the frozen starting point. Players were eligible for the baseline cohort with at least 50 at-bats. The later September 26, 2026 population file was used as the endpoint.

Forward-period counting statistics were derived by subtracting each player's September 2 cumulative totals from his September 26 totals. This prevents the forward analysis from simply comparing two season-to-date classifications. Players were required to record at least 20 forward-period at-bats to remain in the final analysis.

Duplicate player names were matched using player and team information where necessary. No Fantasy State thresholds were changed after the forward outcomes were calculated.

3. Forward-Period Calculations

The forward-only counting statistics were used to reconstruct batting average, on-base percentage, slugging percentage, strikeout rate, and walk rate. These five metrics were then processed through the same Overall and XP formulas used by the Batter Analyzer.

Forward Expected XP was calculated from forward Overall, and Forward XP Divergence was calculated as:

$$\text{Forward XP Divergence} = (\text{Forward XP} - \text{Forward Expected XP}) / \text{Forward Expected XP}$$

4. Study Population

The frozen September 2 snapshot contained 503 hitters meeting the ≥ 50 AB baseline requirement. After applying the ≥ 20 forward-AB requirement, 348 hitters remained in the final forward-performance sample.

5. Fantasy State Results

Frozen Fantasy State	N	Mean Forward XP Divergence	Median
Strong	21	-0.58%	+1.48%
Stable	192	-6.17%	-2.20%
Vulnerable	58	-7.01%	-4.90%
High Risk	77	-12.30%	-4.98%

The mean forward results followed the ordinal structure of Fantasy State: Strong produced the least-negative forward divergence, followed by Stable, Vulnerable, and High Risk. The separation was not perfectly monotonic in the medians, but the mean results moved in the expected direction across all four classifications.

6. Stable vs. Warning-State Analysis

For the primary grouped comparison, Stable players were compared with the combined Vulnerable + High Risk warning group. Strong players were not included in this comparison.

Stable: -6.17% Vulnerable + High Risk: -10.03%

Observed separation: 3.86 percentage points

7. Permutation Test

A two-sided permutation test was used to evaluate whether a Stable-versus-warning separation at least as large as the observed result could arise under random reassignment of the group labels. Using 200,000 random permutations with the observed group sizes preserved, the estimated two-sided p-value was approximately 0.035.

This result provides evidence of an association in this observation window, but it should not be interpreted as proof that Fantasy State is a validated forecasting model. The effect should be tested across additional independent dates, seasons, and samples.

Research Question

Do Fantasy State classifications established at the beginning of an observation window correspond to differences in subsequent XP Divergence?

Interpretation

In this 348-player sample, the frozen Fantasy State classifications were associated with different subsequent XP Divergence outcomes. Mean forward divergence progressed in the model's expected ordinal direction from Strong to Stable to Vulnerable to High Risk.

Stable players also produced less-negative forward XP Divergence than the combined Vulnerable + High Risk group by 3.86 percentage points. The permutation analysis indicates that a separation of this magnitude was uncommon under random reassignment of the observed outcomes.

These findings do not establish that Fantasy State predicts future player performance. They show that, during this observation window, the statistical conditions identified by Fantasy State corresponded to meaningfully different forward outcomes.

Limitations

This analysis represents one late-season observation window. Playing time, injuries, roster changes, schedule effects, and the short forward sample can influence individual results. The study also evaluates an internally constructed outcome—XP Divergence—using variables derived from the same five batting metrics that underpin Fantasy State. Replication is therefore necessary before making broader forecasting claims.

Future Analysis

The next step is to repeat the same frozen-snapshot methodology across additional independent windows and seasons without changing the Fantasy State thresholds. Repeated tests can determine whether the observed ordering and Stable-versus-warning separation persist, weaken, or disappear across different baseball environments.